Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Introduction to

Object Oriented Programming (OOP)

Perkenalan dari mahasiswa

Sebutkan:
(1) Nama lengkap & Jurusan
(2) Daerah asal
(3) Cita-cita (Contoh: programmer, web designer, *etc.*)
(4) Advanced teknologi yang kamu suka? Mengapa?
(4) Ekspektasi (*harapan*) anda, pada saat/sesudah mengambil class ini?
(5) Sudah/belum pernah ambil class saya?

Welcome to *the* Object Oriented Programming *course*

1 - Course Information & Overview

Object-oriented Programming Course

Object-oriented programming (OOP) is a style of programming characterized by the identification of classes of objects closely linked with the methods (*functions*) with which they are associated. It also includes ideas of inheritance of attributes and methods.

Instructor Info & Contact

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Instructor information
- **Office** : Universitas Klabat
- **Mobile phone** : 0811-8432-447
- **Office Hour** : individual appointment

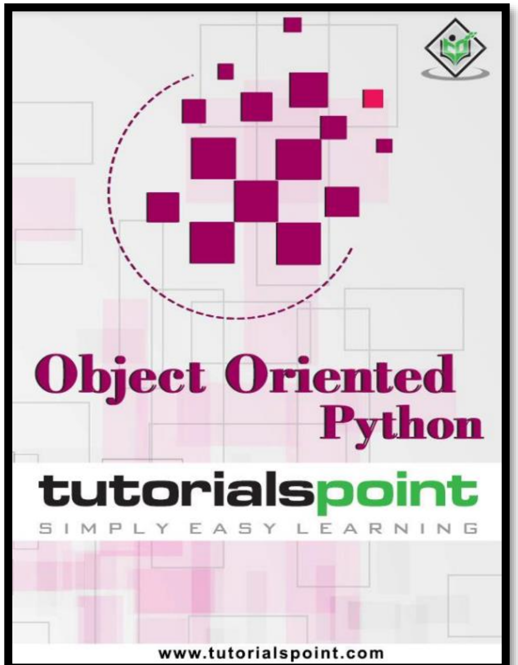
Student Conduct Code
(*framework for student behavior*)

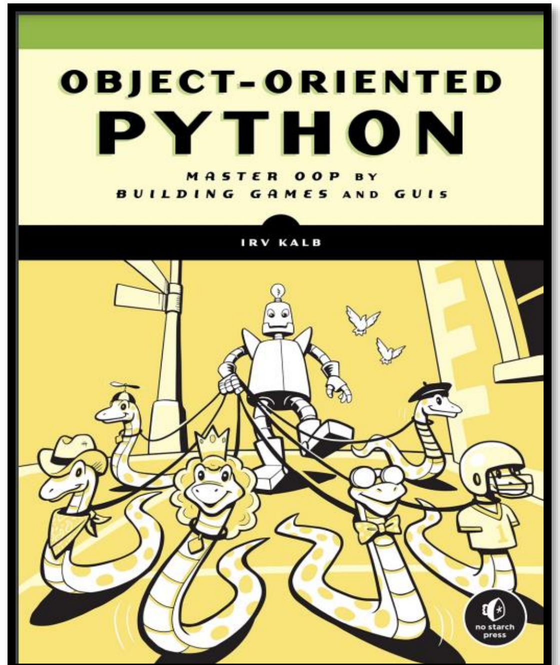
- ✓ Students will practice high standards of professional *honesty* or *integrity* (*quality of being honest and having strong moral principles*).

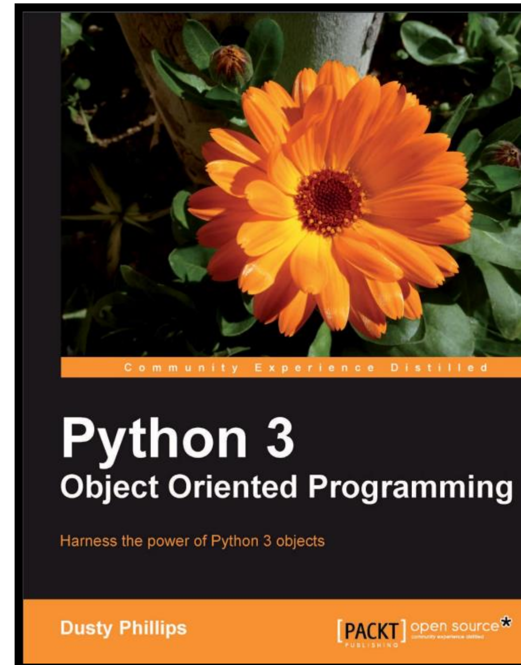
- ✓ In particular, all work (*homework* and *exams*) is to be ***your own individual work***.

- ✓ You may discuss programs but NOT *copy* any code, file or document from any source (*friend* and *online repository*).

Course Textbook & Recommended Readings







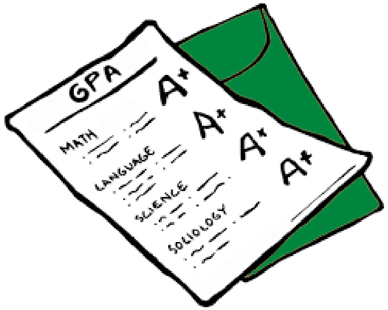
Object oriented python TutorialsPoint Simply Easy Learning. www.tutorialspoint.com.

Irv Kalb. (2022). ***OBJECT-ORIENTED PYTHON Master OOP by Building Games and GUIs***. William Pollock.

Dusty Philips. (2010). ***Computer vision: algorithms and applications***. Packt Publishing.

Grading & Evaluation

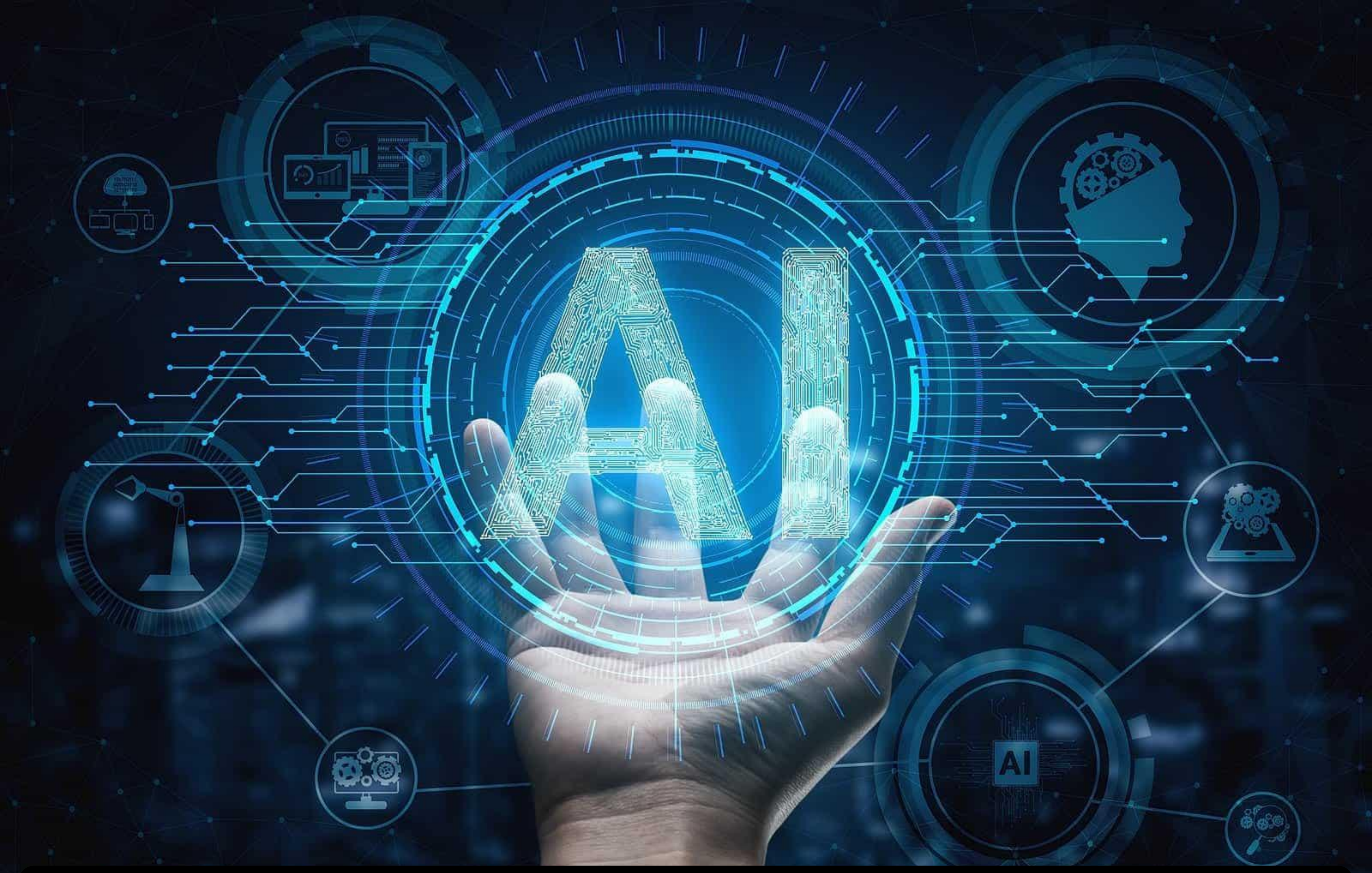
We will have *mid* and *final exams*, so Grading is based on *mid* score, *final* score, *homework/assignments*, *attendance* and *final project* only.



- **Grading**
  - **Mid-term exam**: 20%
  - **Final-term exam**: 20%
  - **Homework/assignments**: 15%
  - **Class activity (1-5)**: 25%
    - Programming assignments
  - **Quiz:** 10%
  - **Attendance**: 10%
  - ***Bonus***: Class participation and Student Conduct ~~(*professional behavior & honesty*): 5%~~

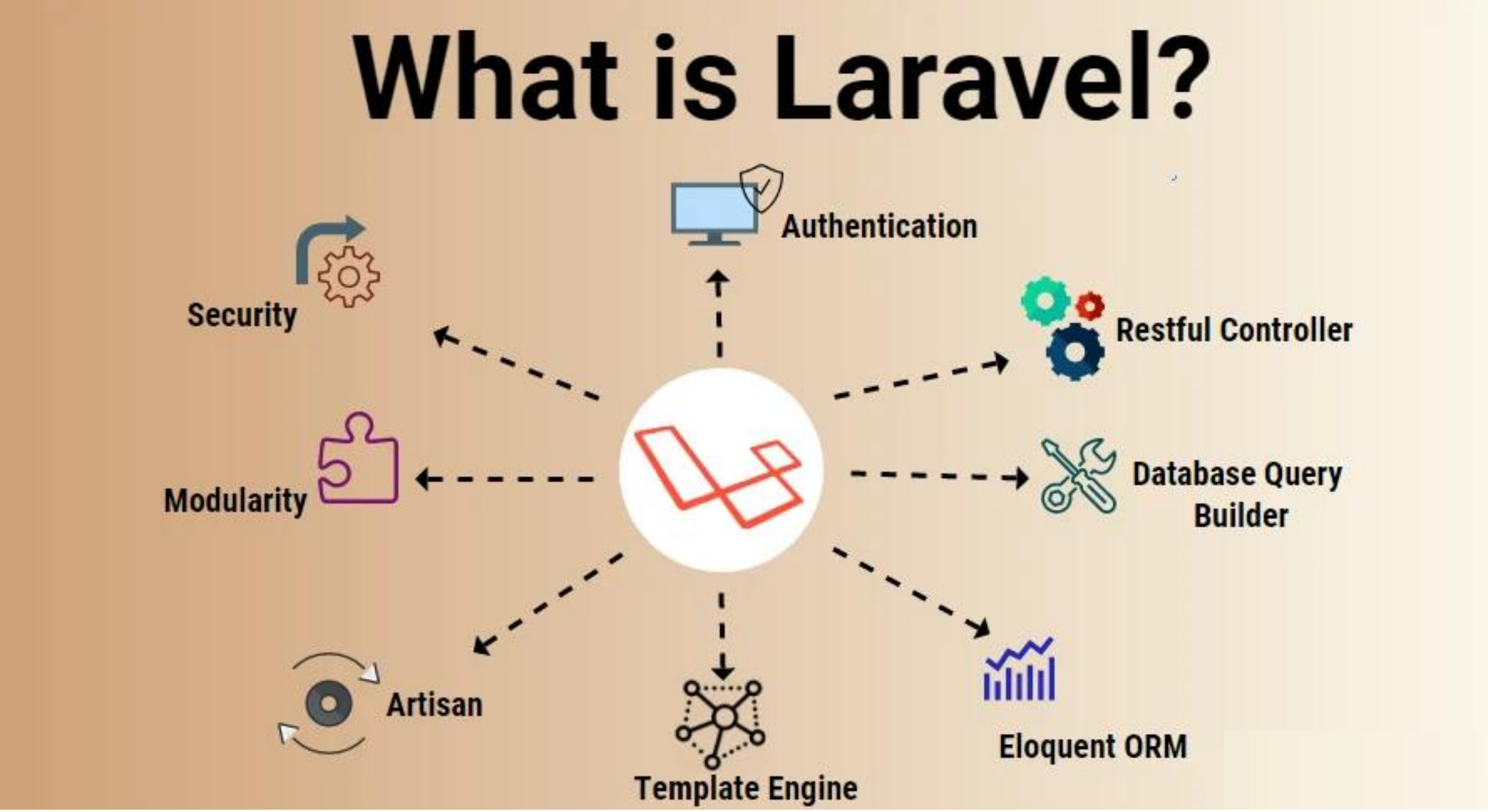
- **Important Notes**
  - You may ***fail*** this course if you miss (*more than*) 7 classes of the whole classes.
  - Academic dishonesty (e.g. *cheating*, *plagiarism*, and *etc.*) will be taken seriously and heavy penalty can be imposed.

2 - What Is Object-Oriented Programming?



Mengapa OOP masih relevan di era AI?

Karena model AI tidak hidup sendiri, **AI system adalah bagian dari sistem besar**. OOP bisa digunakan untuk: (a) Model management (Model, Trainer, Evaluator), (b) Data pipeline (Dataset, Preprocessor), (c) Service layer (InferenceService) dan (d) API dan deployment. Bisa di katakana bawah tanpa OOP, sistem AI sulit dirawat dan dikembangkan.



Laravel ⭐ (paling populer)

✓ Full OOP, MVC

✓ Elegant syntax, sangat cocok untuk akademisi & enterprise

Laravel is a prominent open-source PHP web application framework that uses an elegant, expressive syntax to streamline and enhance the web development process. It follows the Model-View-Controller (MVC) architectural pattern and provides a robust ecosystem of tools and libraries designed to make common tasks, like authentication and database management, much easier.

CodeIgniter is a powerful PHP framework with a very small footprint, built for developers who need a simple and elegant toolkit to create full-featured web applications.

codeigniter4/
framework

PHP framework

4	38k	341	135
Contributors	Used by	Stars	Forks

Java is built around OOPs, which helps in organizing code using classes and objects. Key Features of OOP in Java such as (a) Structures code into logical units (classes and objects), (b) Keeps related data and methods together (encapsulation), (c) Makes code modular, reusable and scalable, (d) Prevents unauthorized access to data and (e) Follows the DRY (Don't Repeat Yourself) principle.

What *is* Object Oriented Programming?

- **Object Oriented** means directed towards **objects**. In other words, it means functionally directed towards **modelling objects**. This is one of the many techniques used for **modelling complex systems** by describing a collection of interacting objects via their data and behavior.

- Python, an Object Oriented programming (OOP), is a way of programming that focuses on using **objects** and **classes** to design and build applications.

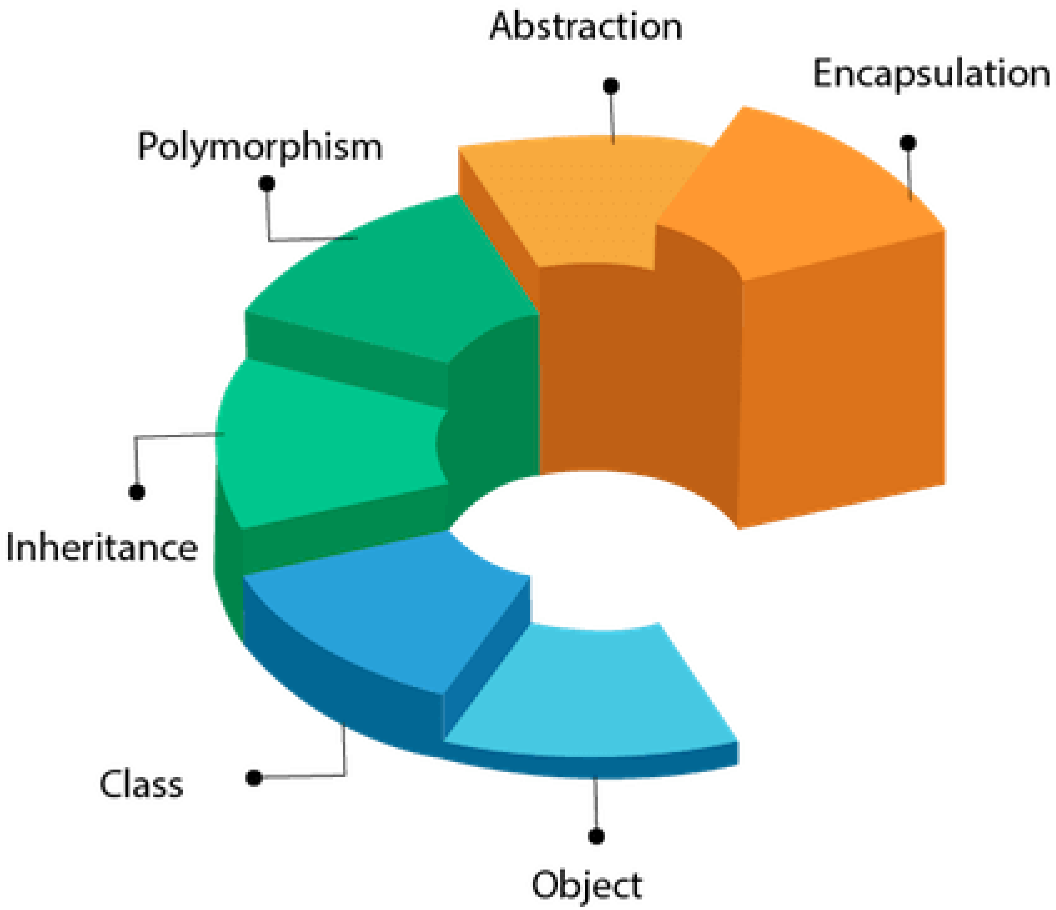
- Major pillars of Object Oriented Programming (OOP) are **Inheritance**, **Polymorphism**, **Abstraction**, and **Encapsulation**.

- Object Oriented Analysis (OOA) is the process of examining a problem, system or task and identifying the objects and interactions between them.

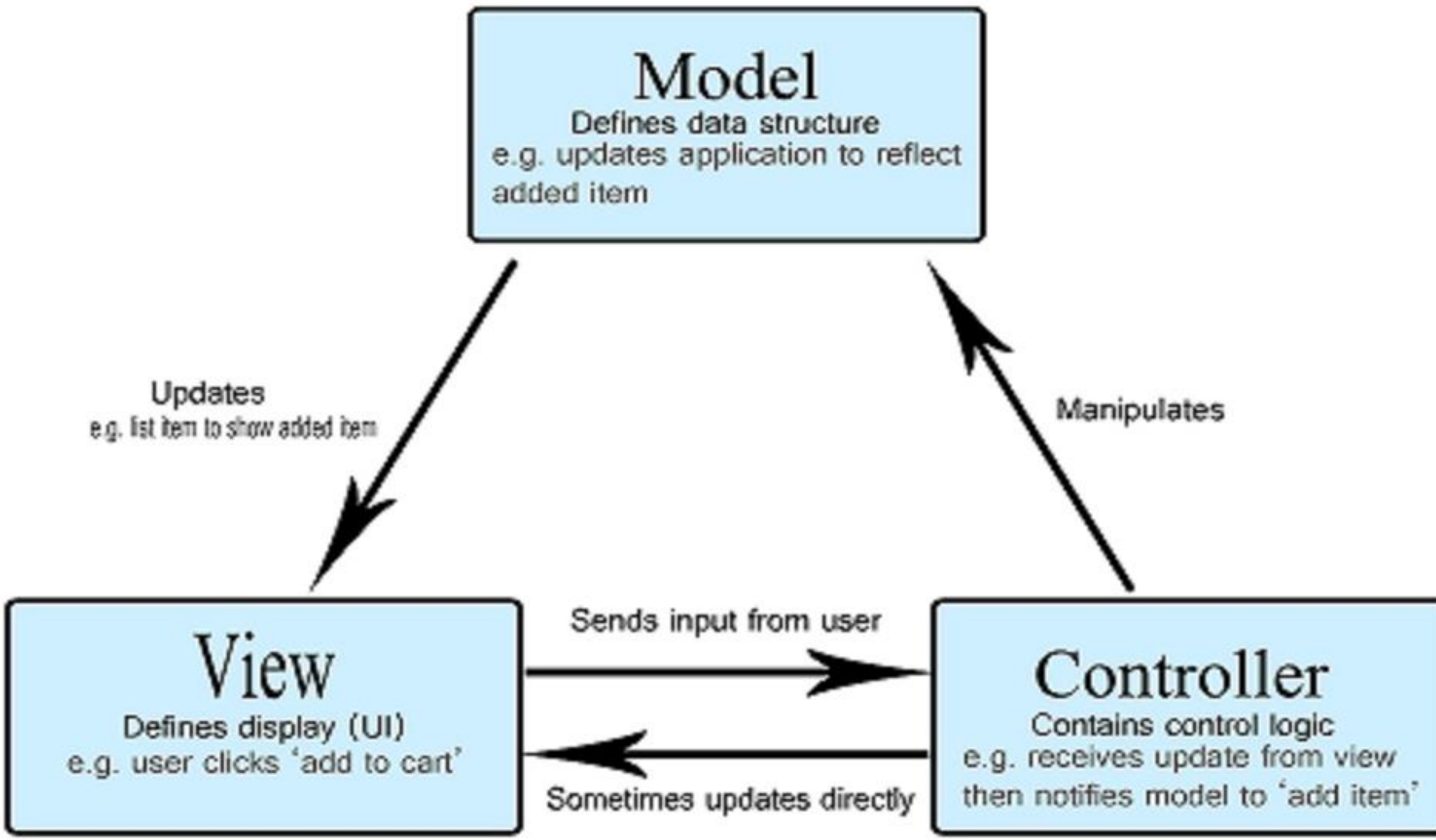
- The ***heart* of Python programming is object and OOP**, however we need not restrict *ourself* to use the OOP by **organizing our code into classes**. OOP adds to the whole design philosophy of Python and encourages a clean and pragmatic way to programming. **OOP also enables in writing bigger and complex programs**.

Object Oriented Programming

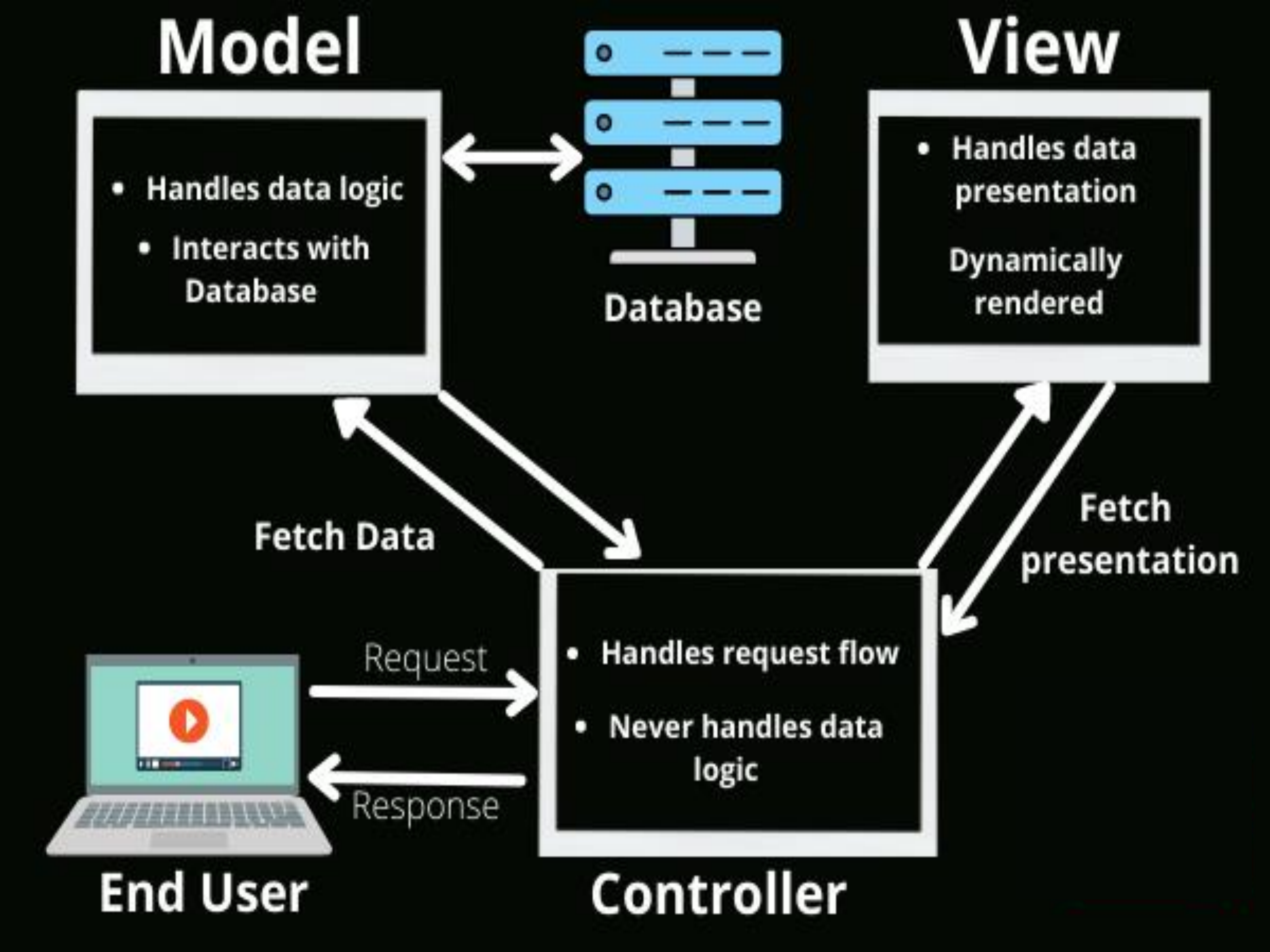
OOPs (Object-Oriented Programming System)



Modelling Complex Systems



MVC (Model View Controller) Design Pattern


Model
Handles data logic
Interacts with Database
Database
View
Handles data presentation
Dynamically rendered
Fetch Data
Fetch presentation
Request
Response
Handles request flow
Never handles data logic
End User
Controller

Jenis-Jenis Web Development

FRONT-END WEB DEVELOPMENT

Tugas utama:

- Para developer pada bagian ini bertugas untuk membangun UI yang dapat membantu pengguna (*users*) untuk mencapai tujuan mereka di dalam situs.
- **Front-end web developer** juga sering ikut dalam aspek perancangan User Experience (UX) dalam project.
- Memiliki *background* di bidang UX dapat menjadi nilai plus bagi seorang front-end developer.

BACK-END WEB DEVELOPMENT

Tugas utama:

- Back-end developer bertugas untuk mengelola server situs web, program (*database*), dan software, dengan tujuan agar semua fitur tersebut dapat bekerja dengan baik.
- **Back-end developer** bekerja dalam sistem seperti ***server***, *sistem operasi*, **API**, dan *basis data*.
- Bertugas untuk mengelola *code* dalam situs untuk keperluan **keamanan**, *konten*, dan *struktur situs*.

Why Choose ObjectOriented Programming?

Python was designed with an object-oriented approach. OOP offers the following advantages:

- ❖ Provides a **clear program structure**, which makes it easy to map real world problems and their solutions.

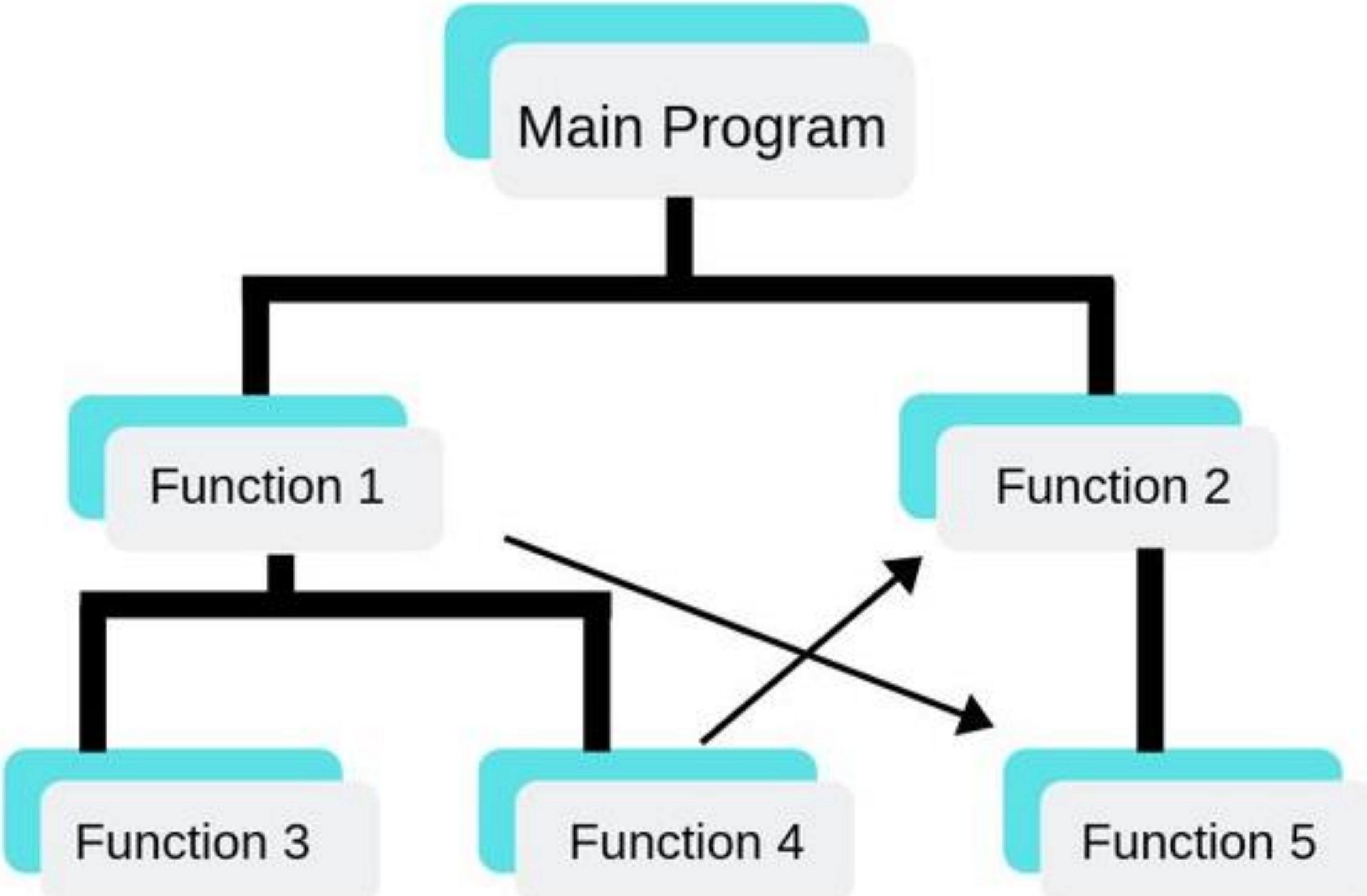
- ❖ Facilitates **easy maintenance and modification** of existing code.

- ❖ Enhances **program modularity** (*program besar dibagi menjadi beberapa bagian program yang lebih kecil*) because each object exists independently and new features can be added easily without disturbing the existing ones.

- ❖ Presents a **good framework** for code libraries where supplied components can be **easily adapted and modified** by the programmer.

- ❖ Imparts/Lets code reusability (*reuse*).

Structured Programming Language



Common Programming Paradigm

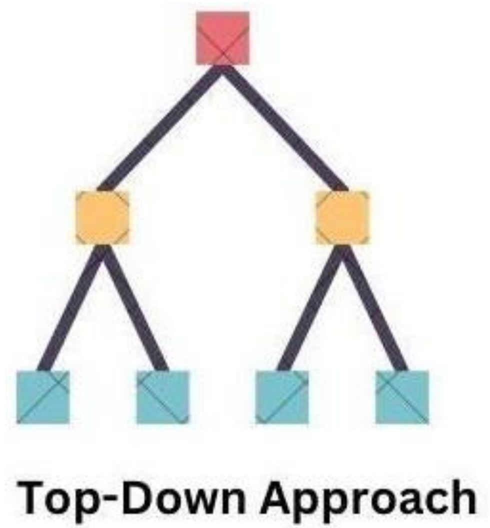
Paradigma dalam dunia pemrograman merupakan sebuah "***cara***" maupun "***sudut pandang***" dari **<u>seorang proramme</u>r** dalam **<u>memecahkan sebuah masalah</u>** berdasarkan bidang pemrograman itu sendiri (Java, Python, C++, Javascript, *etc.*).

Structured/Procedural Programming	Object Oriented Programming
Code is divided into modules or functions	Code is made up of classes and objects
"Top-down" approach	Bottom-up approach
Difficult to modify / manage	Easy to modify / manage
Main function calls other functions	Objects communicate by passing messages
Data is not secured	Data is secure
Less reusability of code	More reusability of code
Less flexibility and abstraction	More flexibility and abstraction

“Top-down" *vs* “Bottom-up" Approach

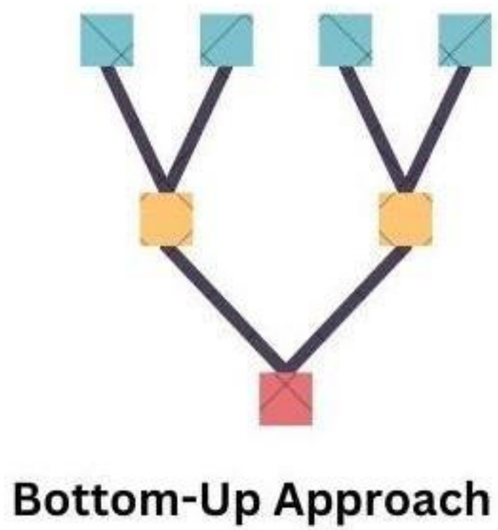
In the beginning, a conceptual or general design of the system is created. Each component is then further divided into sub-components to the desired level of detail.

- ❖ Dimulai dari desain umum dan bergerak ke detail. Fokus pada pandangan sistem secara keseluruhan.
- ❖ Cocok untuk proyek-proyek besar dengan struktur hierarki yang jelas.



In the beginning, focus is put on creating small components or independent modules. These modules are then combined into larger modules, and this process continues until a complete system is formed.

- ❖ Fokus pada pengembangan modul-modul kecil yang dapat diuji secara independen.
- ❖ Cocok untuk pengembangan iteratif atau ketika ada komponen yang sudah ada.



Procedural *vs* Object Oriented Programming

Procedural based programming is derived from structural programming based on the concepts of functions/procedure/routines. It is easy to access and change the data in procedural oriented programming.

On the other hand, **Object Oriented Programming (OOP) allows decomposition of a problem into a number of units called objects and then build the data and functions around these objects.** It emphasis more on the data than procedure or functions. Also in OOP, data is hidden and cannot be accessed by external procedure.

Principles *of* Object Oriented Programming

Object Oriented Programming (OOP) is based on the concept of **objects** rather than **actions**, and **data** rather than **logic**. In order for a programming language to be object-oriented, it should have a mechanism to **enable working with classes** and **objects** as well as the implementation and usage of the fundamental object-oriented principles and concepts namely **inheritance**, **abstraction**, **encapsulation** and **polymorphism**.

Pillars *of* Object-oriented Programming

Encapsulation

This property hides unnecessary details and makes it easier to manage the program structure. Each object's implementation and state are hidden behind well-defined boundaries and that provides a clean and simple interface for working with them. One way to accomplish this is by making the data private.

Inheritance

Inheritance, also called generalization, allows us to capture a hierarchal relationship between classes and objects. For instance, a 'fruit' is a generalization of 'orange'. Inheritance is very useful from a code reuse perspective.

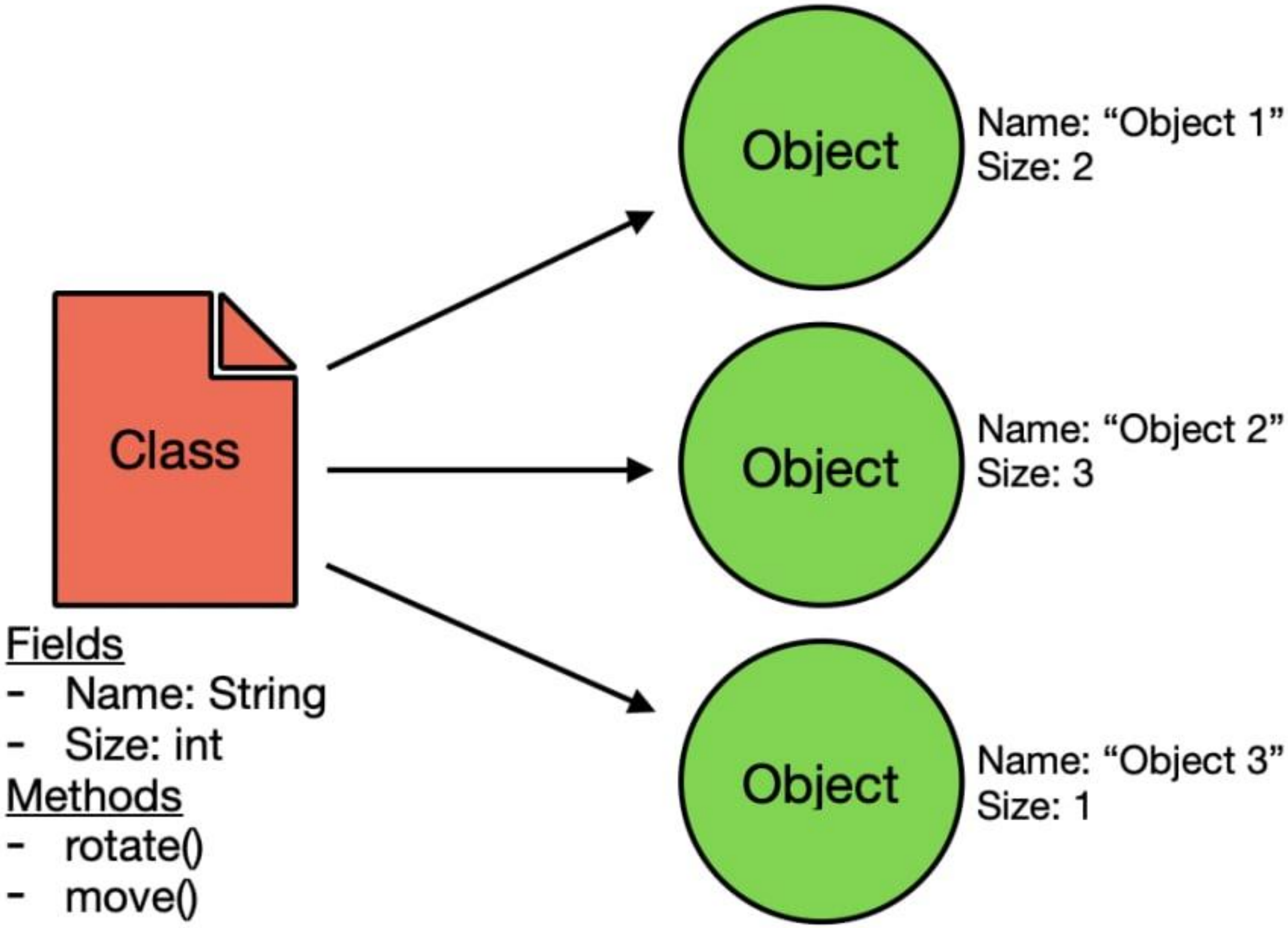
Abstraction

This property allows us to hide the details and expose only the essential features of a concept or object. For example, a person driving a scooter knows that on pressing a horn, sound is emitted, but he has no idea about how the sound is actually generated on pressing the horn.

Polymorphism

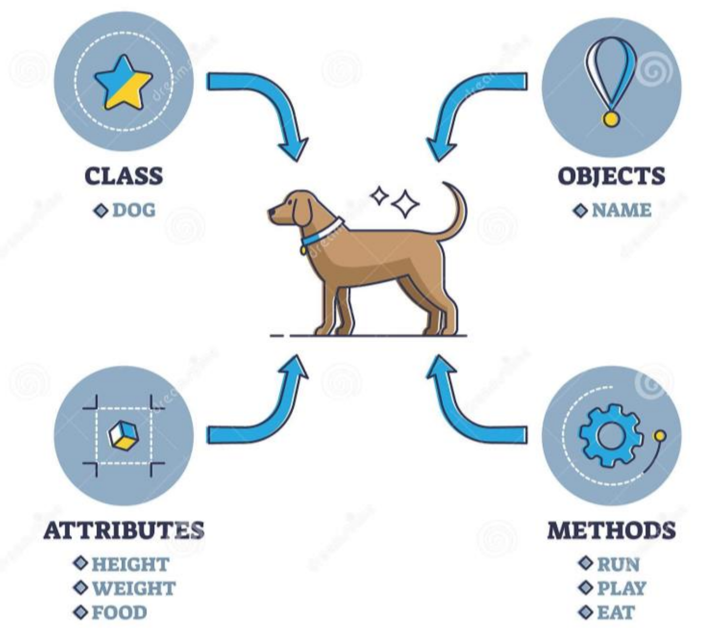
Poly-morphism means many forms. That is, a thing or action is present in different forms or ways. One good example of polymorphism is constructor overloading in classes.

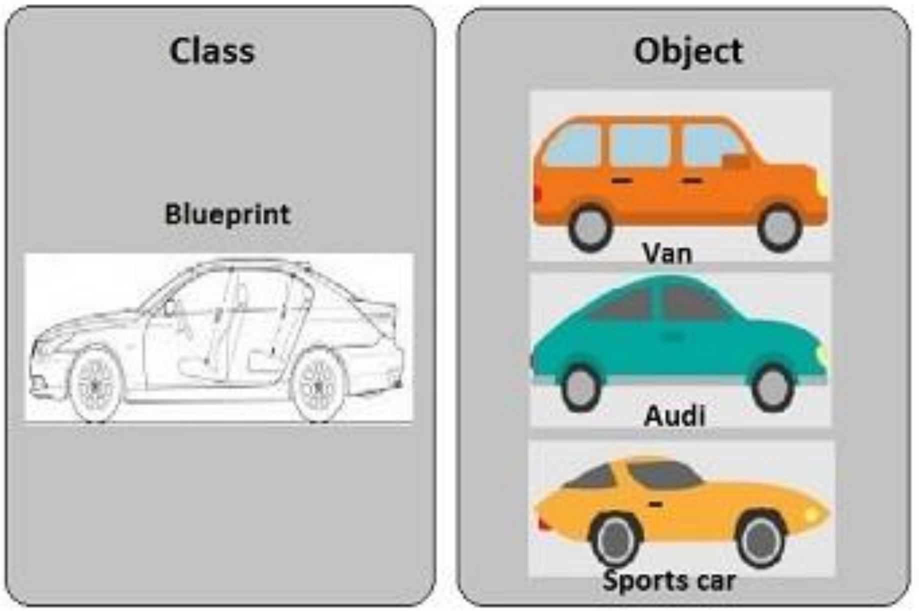
OOP Example



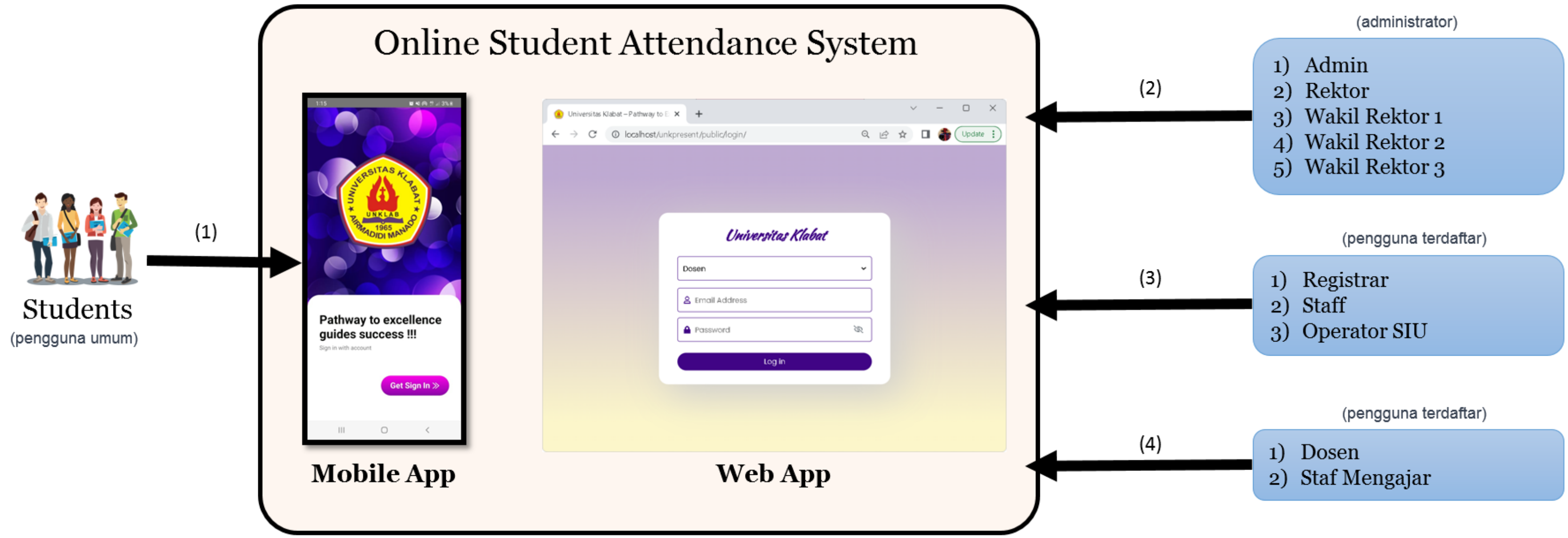
**Each object has its own values for fields and can call the methods found in the class*

OOP Example



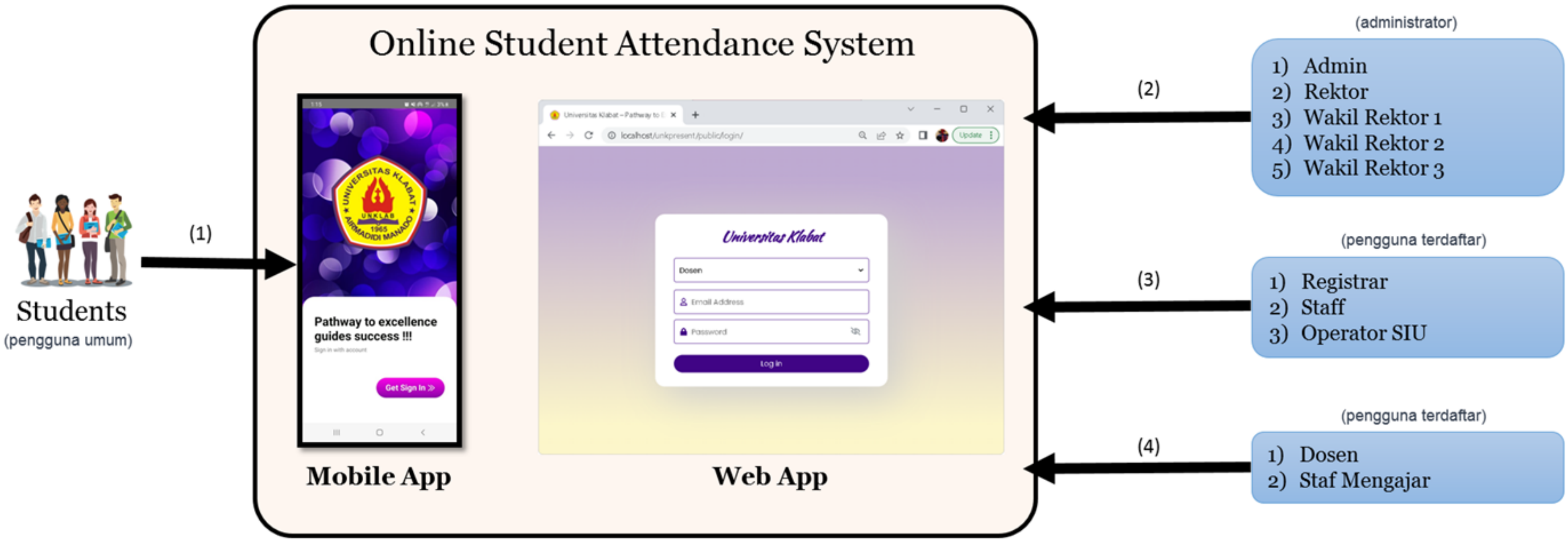


Class adalah "**prototype**" atau "**blueprint**" atau "**rancangan**" yang mendefinisikan variable dan method-methode pada seluruh objek tertentu. Class berfungsi untuk menampung isi dari program yang akan di jalankan, di dalamnya berisi atribut / type data dan method untuk menjalankan suatu program.



1) **Pengguna umum** hanya memiliki akses terbatas ke fitur dan informasi dalam sistem.
2) **Pengguna terdaftar** memiliki hak akses lebih seperti mengunggah atau mengedit.
3) **Administrator** memiliki hak akses penuh dan dapat mengatur dan mengelola sistem secara keseluruhan.

Sistem presensi mahasiswa di UNKLAB, dikembangkan dengan mengadopsi Framework MVC (AIMVC) yang dibuat di UNKLAB dan menggunakan pendekatan OOP.



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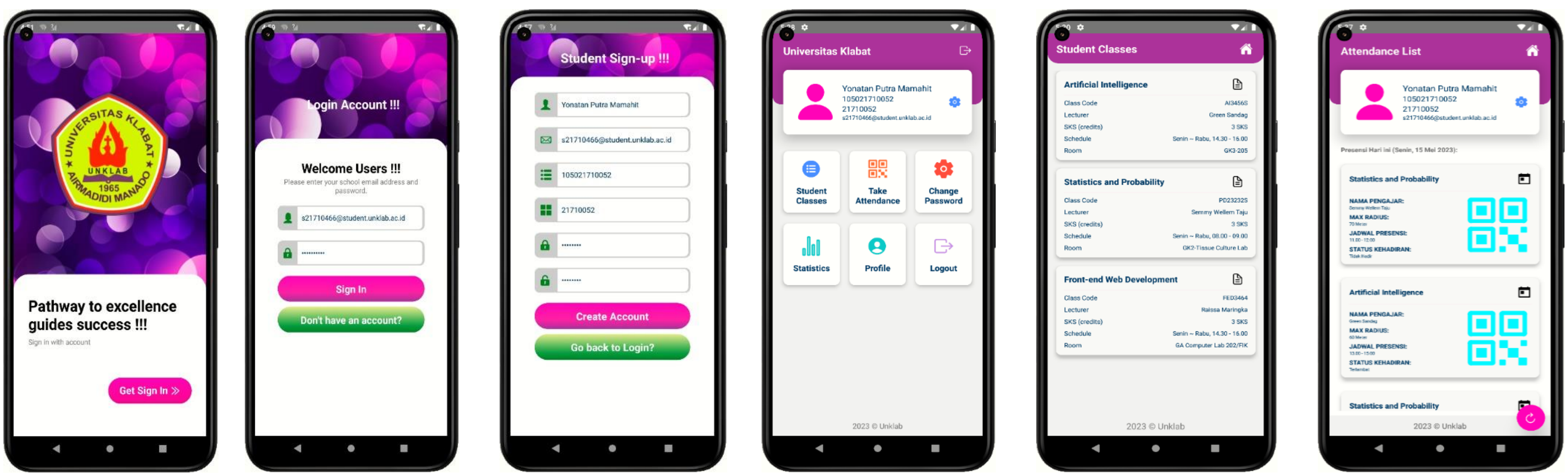

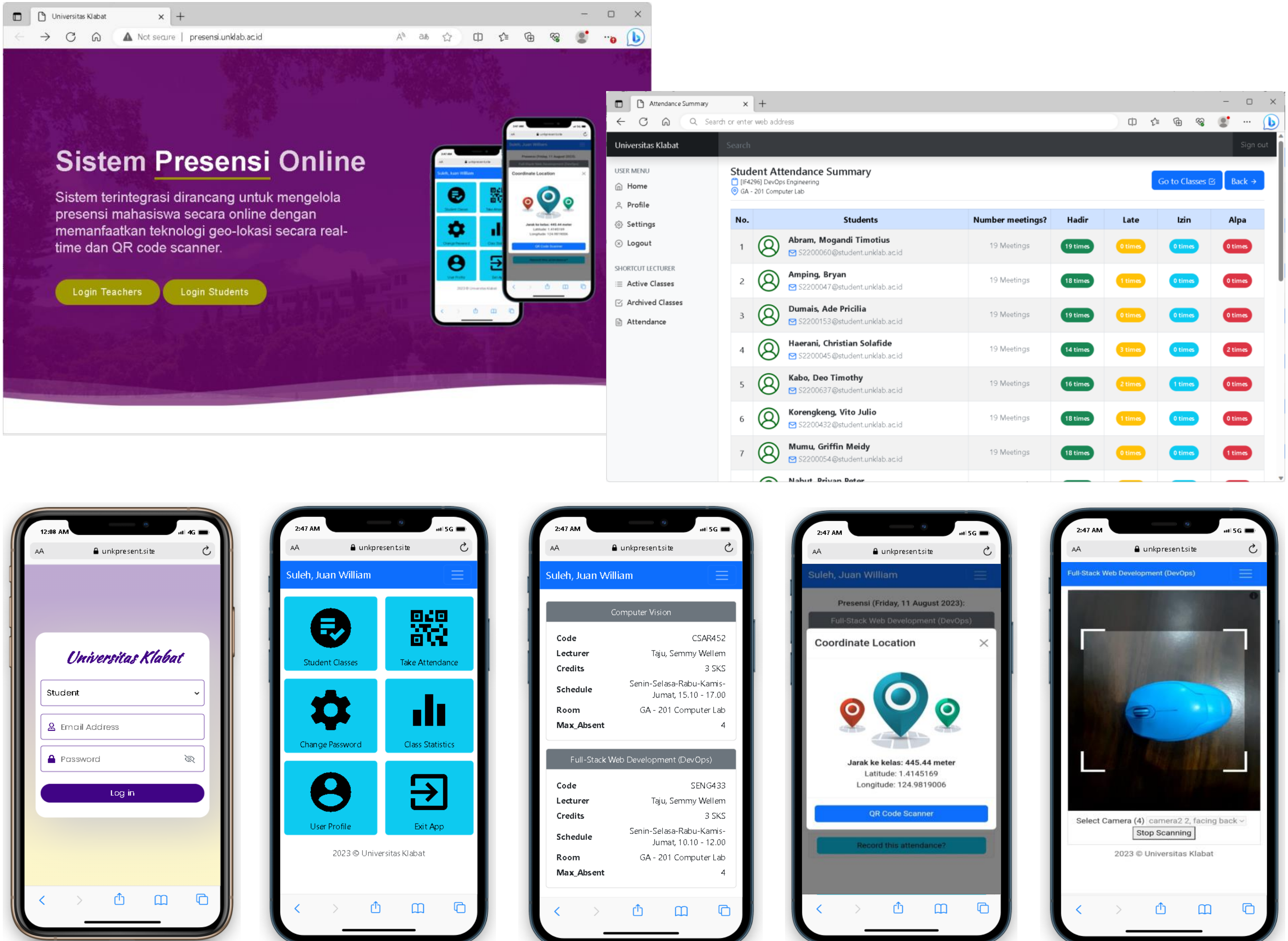
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Sistem Presensi Online
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Sign out
USER MENU
Home
Profile
Settings
Logout
SHORTCUT LECTURER
Active Classes
Archived Classes
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Amping, Bryan
Dumais, Ade Pricilia
Haerani, Christian Solafide
Kabo, Deo Timothy
Korengkeng, Vito Julio
Mumu, Griffin Meidy
19 Meetings
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Password
Log in
Suleh, Juan William
Student Classes
Take Attendance
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Class Statistics
User Profile
Exit App
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Computer Vision
Code
CSAR452
Lecturer
Taju, Semmy Wellem
Credits
3 SKS
Schedule
Senin-Selasa-Rabu-Kamis-Jumat, 15.10 - 17.00
Room
GA - 201 Computer Lab
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Full-Stack Web Development (DevOps)
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SENG433
Lecturer
Taju, Semmy Wellem
Credits
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Schedule
Senin-Selasa-Rabu-Kamis-Jumat, 10.10 - 12.00
Room
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Presensi (Friday, 11 August 2023):
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Jarak ke kelas: 445.44 meter
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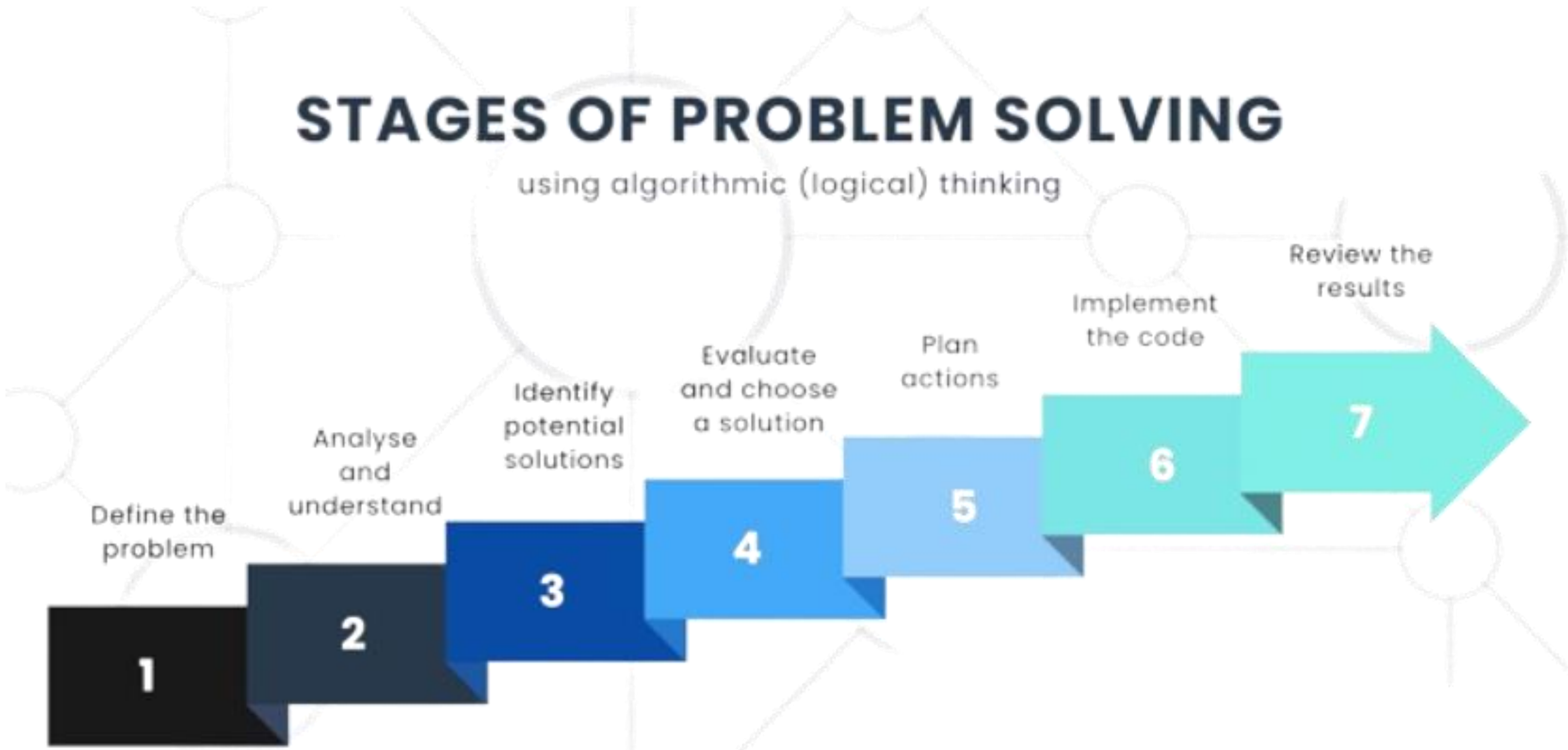
Bagaimana cara untuk dapat mengimplementasikan konsep Object-Oriented Programming (OOP) secara efektif dalam pengembangan sistem presensi mahasiswa di UNKLAB?



Exercise *for* Students

Programming paradigms *in* Python

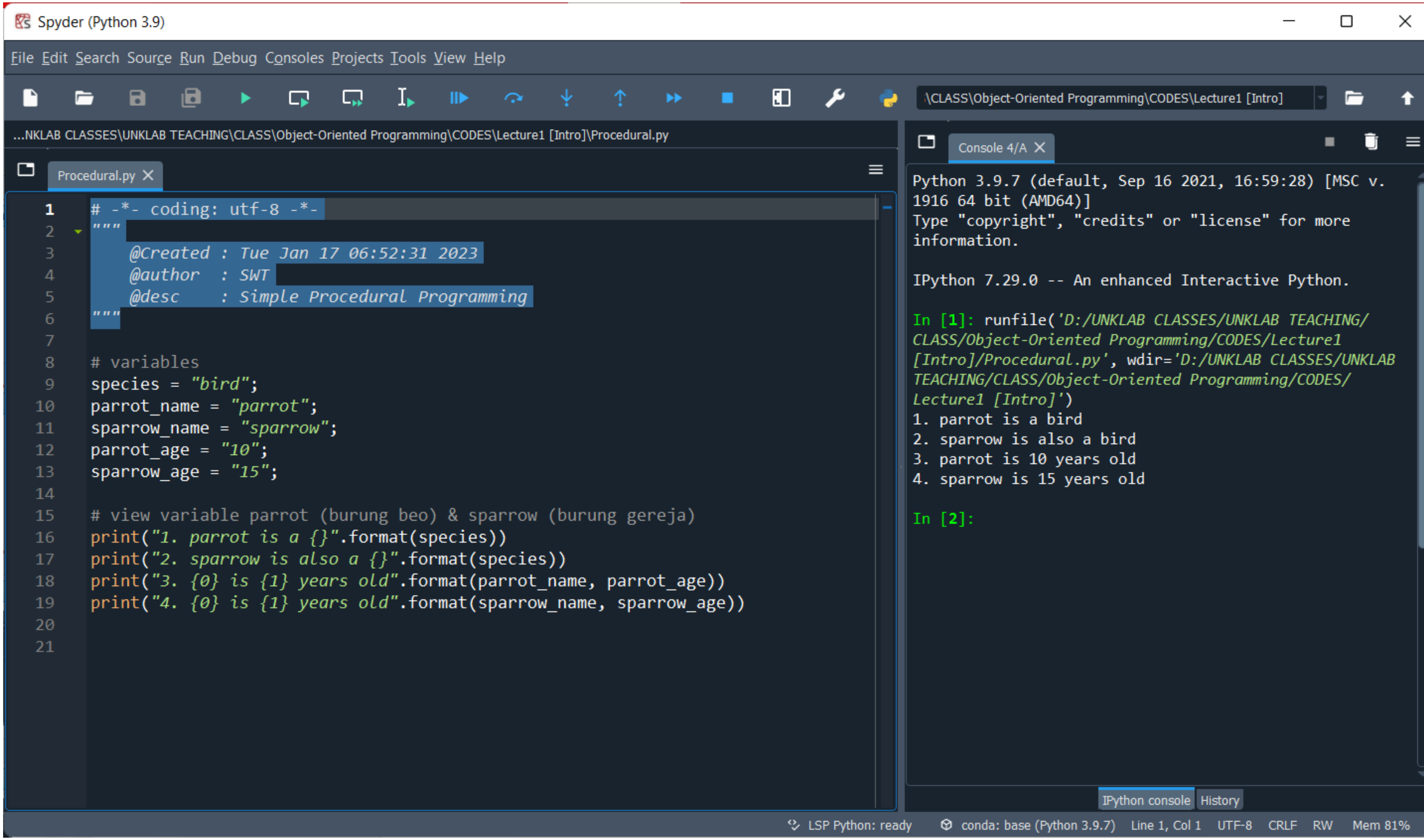
There are three programming paradigms that are supported by Python:

1. Procedural programming
2. Object-Oriented Programming
3. Functional programming

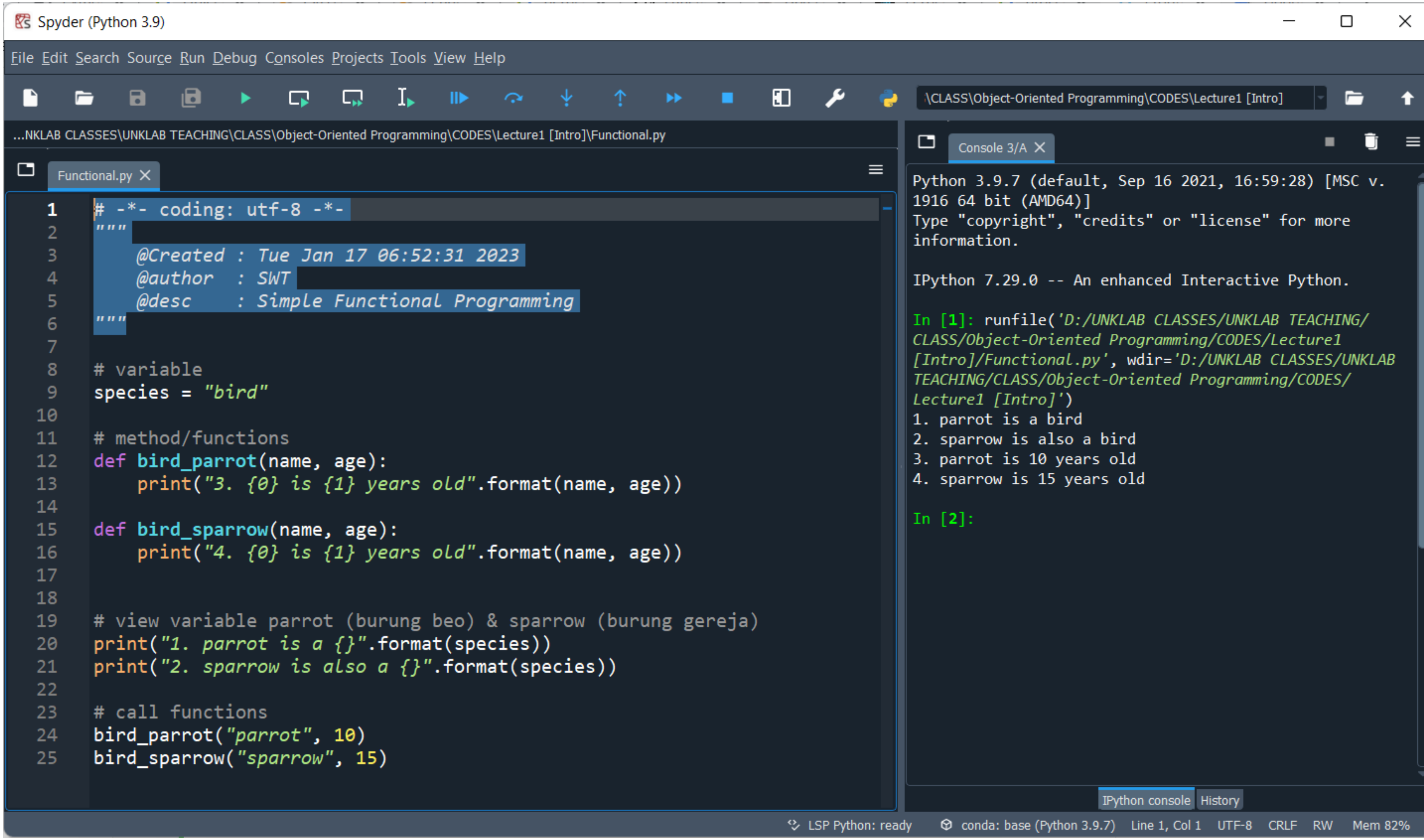
A *programming paradigm* is a framework or basic model (*pattern*) that guides the approach and methods in writing program code. Programming paradigm refers to the ***basic style*** or the <u>way of writing program code</u>.

<u>Paradigma pemrograman</u> adalah *pola pemikiran*, *kerangka berpikir*, model dasar, *perspektif*, ataupun gaya (*style*) code yang membimbing pendekatan dalam penulisan kode program. Masing-masing kita di dalam class ini akan memiliki *style* sendiri dalam penulisan code program. Menurut kalian, apakah mudah untuk menyatukan cara berpikir dalam pemrograman? Khususnya dalam penulisan code program?

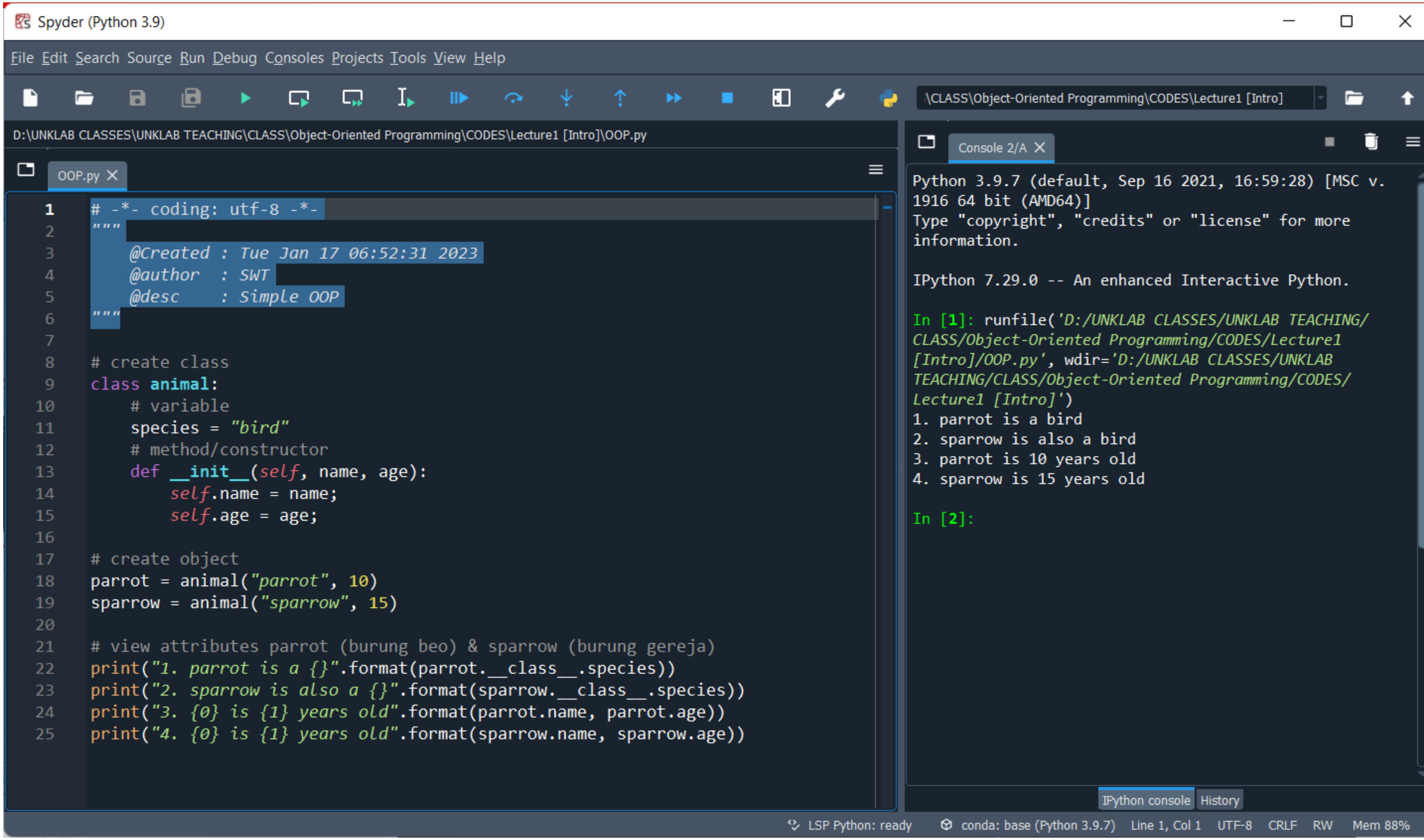
Procedural Programming



Functional Programming



Object-Oriented Programming



Test your understanding

The following are some questions to reinforce your understanding of the concepts we've covered so far.






Question #1

Buatlah dengan kata-kata anda sendiri, definisi dari OOP, Class dan juga object?

Question #2

Sebutkan 4 major pillars of Object Oriented Programming (OOP) ???

Question #3

Jelaskan dengan singkat perbedaan dari tiga tipe programming paradigms yang di supported oleh bahasa pemrograman Python (Procedural programming, Object-Oriented Programming, Functional programming) ???

END PRESENTATION

Thank you for your attention

Instructor: S – W – T